



RKDF UNIVERSITY, RANCHI

M.Sc. (Bio-Technology)

SCHEME SEMESTER – I

S.No.	Subject Code	Subject Name	Marks Distribution				
			Internal	External		Total	
				Max	Min	Max	Min
1.	PBT101	Molecular Biology & Genomics	30	70	21	100	35
2.	PBT102	Cell Biology	30	70	21	100	35
3.	PBT103	Microbiology	30	70	21	100	35
4.	PBT104	Industrial Biotechnology	30	70	21	100	35
Total						400	
<i>Practical</i>							
S.No.	Code	Subject Name	Max		Min		
1.	PBT155	Practical - I Lab	100		50		



RKDF UNIVERSITY, RANCHI

M.Sc. (Bio-Technology)

Choice Based Credit System Semester - I

S.No.	Subject Code	Subject Name	L	T	P	Credit
1.	PBT101	Molecular Biology & Genomics	3	1	0	4
2.	PBT102	Cell Biology	3	1	0	4
3.	PBT103	Microbiology	3	1	0	4
4.	PBT104	Industrial Biotechnology	3	1	0	4
5.	PBT155	Practical - I Lab	0	0	8	4
Total			12	4	8	20



RKDF UNIVERSITY, RANCHI

M.Sc. (Bio-Technology)

SEMESTER - I

Branch	Subject Title	Subject Code
M.Sc. Biotechnology	Molecular Biology& Genomics	PBT101

Group – A

1. Introduction to Molecular Biology and Genetics.

Genome organization Organization of bacterial genome; Structure of eukaryotic chromosomes; Role of nuclear matrix in chromosome organization and function; Matrix binding proteins; Heterochromatin and Euchromatin; DNA reassociation kinetics (Cot curve analysis); Repetitive and unique sequences; Satellite DNA; DNA melting and buoyant density; Nucleosome phasing; DNase I hypersensitive regions; DNA methylation & Imprinting.

2. DNA Structure; Replication; Repair & Recombination

Structure of DNA-A-, B-, Z- and triplex DNA; Measurement of properties-Spectrophotometric, CD, AFM and Electron microscope analysis of DNA structure; Replication initiation, elongation and termination in prokaryotes and eukaryotes; Enzymes and accessory proteins; Fidelity; Replication of single stranded circular DNA; Gene stability and DNA repair- enzymes; Photoreactivation; Nucleotide excision repair; Mismatch correction; SOS repair; Recombination: Homologous and non-homologous; Site specific recombination; Chi sequences in prokaryotes; Gene targeting; Gene disruption; FLP/FRT and Cre/Lox recombination.

3. Insertion elements & Transposons.

4. Gene, mutation and mutagenesis: UV and chemical mutagens; types of mutation; Ames test for mutagenesis; Methods of genetic analysis, Strain improvement and Mutator gene.

Group – B

1. Transcription: Translation machinery; Ribosomes; Composition and assembly; Universal genetic code; Degeneracy of codons; Termination codons; Isoaccepting tRNA; Wobble hypothesis; Mechanism of initiation, elongation and termination; Co- and post-translational modifications; Genetic code in mitochondria; Transport of proteins and molecular chaperones; Protein stability; Protein turnover and degradation Prokaryotic transcription, Eukaryotic transcription, RNA polymerase, General and specific transcription factors, Regulatory elements and mechanisms of transcription regulation, Transcriptional and post-transcriptional gene silencing.
2. Modifications in RNA: 5- Cap formation, Transcription termination, 3- end processing and polyadenylation, splicing, Editing, Nuclear export of m RNA, m RNA stability.



RKDF UNIVERSITY, RANCHI

M.Sc. (Bio-Technology)

3. Translation: Prokaryotic and eukaryotic translation, co- and post- translation modifications of proteins.

Group - C

1. Bacterial genetic system: Transformation, Conjugation, Transduction, Bacterial genetics map with reference to E.coli.
2. Biology for Cancer: Oncogenes and tumour suppressor genes; Viral and cellular oncogenes, tumour suppressor genes from humans.
3. Antisense and Ribozyme technology: Molecular mechanism of antisense molecules, applications of antisense and ribozyme technologies.
4. Holiday junction, gene targeting gene disruption, Rec A and other recombinases.

Group - D

1. Mapping of Genome: Genetic and physical maps, Physical mapping and map- based cloning, Southern and fluorescence in situ hybridization (FISH) for genome analysis micro array analysis.
2. Genome sequencing: Genome sizes, organelle genomics, Genomic libraries YAC, BAC, libraries, Strategies for sequencing genome.
3. Mendelian Genetics

Introduction to human genetics; Background and history; Types of genetic diseases; Role of genetics in medicine; Human pedigrees; Patterns of single gene inheritance- autosomal recessive; Autosomal dominant; X linked inheritance; Complicating factors - incomplete penetrance; variable expression; Multiple alleles; Co dominance; Sex influenced expression; Hemoglobinopathies - Genetic disorders of hemoglobin and their diseases. Non Mendelian inheritance patterns Mitochondrial inheritance; Genomic imprinting; Lyon hypothesis; isodisomy; Complex inheritance-genetic and environmental variation; Heritability; Twin studies; Behavioral traits; Analysis of quantitative and qualitative traits.

Essential Readings

- Garder, Principles of genetics, Wiley Publications, 8th edition
- Levin, Gene VI to Gene VIII, Oxford Pub.
- Friefelder, Essentials of Molecular Biology, Panima Pub
- T. A. Brown, Genome-2 2nd Edition
- Old & primrose, Principle of Gene Manipulation, Blackwell Pub.
- Weaver Molecular Biology, Mc Graw Hill
- Brown, Gene Cloning and DNA analysis, Blackwell Pub.
- Winnacker, From genes to clones, Panima Pub.
- P.C. Tumer, Instant notes in Immunology, Viva books Pub.



RKDF UNIVERSITY, RANCHI

M.Sc. (Bio-Technology)

- Griffith, Introduction to genetic analysis, Freeman publication, 8th edition
 - Robert Brooker, Genetics, McGraw-Hill
 - Strickberger, Genetics, Prentice Hall Pub.
 - T. A. Brown, Gene Cloning DNA analysis- Blackwell Pub.
 - Stephen Hunt, Functional Genomics Oxford, Tokyo
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RKDF UNIVERSITY, RANCHI

M.Sc. (Bio-Technology)

SEMESTER - I

Branch	Subject Title	Subject Code
M.Sc. Biotechnology	Cell Biology	PBT102

Group – A

1. Molecular logic of living organism, cell and its biochemical organization. Cell theory.
2. Membrane Structure and Function Structural models; Composition and dynamics; Transport of ions and macromolecules; Pumps, carriers and channels; Endo- and Exocytosis; Membrane carbohydrates and their significance in cellular recognition; Cellular junctions and adhesions; Structure and functional significance of plasmodesmata.
3. Cellular organelles-plasma membrane, cell wall, their structural organization; Mitochondria, chloroplast, nucleus and other organelles and their organization. Nucleus – Structure and function of nuclear envelope, lamina and nucleolus; Macromolecular trafficking; Mitochondria – structure, organization of respiratory chain complexes, ATP synthase, Structure-function relationship; Mitochondrial DNA and male sterility; Origin and evolution; Chloroplast – Structure-function relationship; Chloroplast DNA and its significance; Chloroplast biogenesis; Origin and evolution. Cellular energy transaction-role of mitochondria and chloroplast.

Group – B

1. Transport of nutrients, ions and macromolecules across membranes.
2. Cell motility-Cilia, flagella of Eukaryotes and Prokaryotes.
3. Structure and function of microbodies, Golgi apparatus, Lysosomes and Endoplasmic Reticulum; Organization and role of microtubules and microfilaments; Cell shape and motility; Actin-binding proteins and their significance; Muscle organization and function; Molecular motors; Intermediate filaments; Extracellular matrix in plants and animals

Group – C

1. Cellular responses to environmental signals in plants and animals; mechanisms of signal transduction.
2. Chromatin organization and packaging; Cell cycle and control mechanisms;
3. Cellular basis of differentiation and development - mitosis, meiosis gametogenesis and fertilization.



RKDF UNIVERSITY, RANCHI

M.Sc. (Bio-Technology)

Group – D

1. Microscope and its applications – Light, phase contrast and interference, Fluorescence, Confocal.
2. Electron microscopy (TEM and SEM),
3. Electron tunnelling and Atomic Force Microscopy, etc.

Essential Readings

Alberts. Molecular Biology of cell. GarlandPub

Verma, Cell biology, Genetics, Molecular Biology, Evolution & Ecology. 2006

Karp, Cell & Molecular Biology: concepts & Experiments. 4th Edition.

Lodish, Cell & Molecular Biology, W.H. Freeman. 5th Edn.

Watson, Molecular Biology of the Genes. Pearson Publication.

Wilson, & Walker. 1995. Principles and techniques of practical Biochemistry.

Becker. 1996. Biotechnology: A laboratory course. Alp

Lenhinger. Principles of biochemistry, 2nd Edn.

Glick, Molecular Biotechnology, ASM Publication.

Becker & Hardin, The world of the Cell, Pearson Pub.

C.B. Powar, Cell Biology, Himalaya Press.

Nelson & Cox, Lehninger Principle Biochemistry, Freeman Pub.

Desiker, Cell & Development Biology, Dominant Pub.

Albert, Essential Cell Biology, Garland Science.

Geoffrey Cooper, The- Cell Molecular Approach, ASM Pub.

Ben Hui Liu, Statistical Genomics : Linkage, mapping & QTL Analysis, CRC Press



RKDF UNIVERSITY, RANCHI

M.Sc. (Bio-Technology)

SEMESTER – I

Branch	Subject Title	Subject Code
M.Sc. Biotechnology	Microbiology	PBT103

Group -A

1. History of microbiology; Development of pure culture methods; Enrichment culture methods; Development of microbiology in the twentieth century.
2. Methods of Microbiology: Isolation, pure culture techniques, staining of bacterial cells and its organelles, methods of sterilization-physical and chemical, selection and construction of culture media, enrichment culture technique, assay of amino acids and antibiotics.
3. Microbial evolution, Systematics and Taxonomy: Evolution of earliest life forms, bacterial identification, nomenclature and classification, new approach to bacterial taxonomy / classification including ribotyping and ribosomal RNA sequencing.

Group - B

1. Prokaryotic cells: structure function: General structure and feature; cell wall of eubacteria, flagella, cell inclusions-endospore and gas vesicles.
2. Prokaryotic diversity: Bacteria; Brief account of all groups of bacteria and cyanobacteria, Rickettsias, Chlamydiae, and Mycoplasma.
3. Archae: Archaeobacteria extremophilic microbes - their biotechnological potentials.
4. Viruses: Classification, morphology and composition of virus in general.
5. Bacteriophage: phi X174 cyanophage and retroviruses, viroids and prions.

Group - C

1. Microbial growth and Physiology: The definition of the growth, growth curve, measurement of growth and growth yields, synchronous and continuous growth. culture collection and maintenance of culture. Life style of Prokaryotes, Unicellular Eukaryotes
2. Overview of microbial nutrition: Types and mode of nutrition in bacteria.
3. Metabolic diversity among microorganisms: Photosynthesis in microorganisms, chemolithotrophy, sulphate reduction.
4. Nitrogen metabolism- nitrate reduction, nitrifying and denitrifying bacteria nitrogen fixation nitrogen cycle.

Group - D

1. Secondary metabolites/ Bioactive substances
2. Brief account of toxins of bacteria and cyanobacteria
3. Microbes used as bio-fertilizer



RKDF UNIVERSITY, RANCHI

M.Sc. (Bio-Technology)

Essential Readings

- Pelczar M.J. Chan, 5th Edition, Microbiology
 - Roger Y. Stanier, 5th Edition General microbiology
 - Powar & Dagainawala Vol I & Vol II, General Microbiology
 - Prescott L. M. Microbiology, 6th Edition
 - Atlas R.M. Microbiology
 - Johnson, Laboratory Experiments in Microbiology, 6th Edition, Pearson Education
 - Harold J. Benson, Microbiological applications, 6th Edition
 - Singleton Sainsbury, Dictionary of Microbiology & Molecular Biology, John Wiley
 - R.C. Dubey & Maheshwari, A Textbook of Microbiology, 1st Edn, 2005.
 - Medical Microbiology, Anantnarayan
 - Nicklin, Instant Notes in Microbiology, 2nd Edn.
 - Stanier, General Microbiology, 5th Edn.
 - Ingraham, Introduction to microbiology, 3rd Edn
 - Moat, Microbial Physiology, 4th Edn.
 - Ignacimuthu, Methods in Microbiology
 - Black, Microbiology Principle & Exploration, 6th Edition.
 - Torotora, Microbiology: An Introduction 8th Edition.
 - Cuppuccino, Microbiology : A Laboratory Manual, 8th Edition
 - White, Physiology & Biochemistry of Prokaryotes
 - Alexopoulos, Introductory Mycology
 - Rajvaidya, Applied Microbiology Vol. I to V, APH Pub.
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RKDF UNIVERSITY, RANCHI

M.Sc. (Bio-Technology)

SEMESTER – I

Branch	Subject Title	Subject Code
M.Sc. Biotechnology	Industrial Biotechnology	PBT104

Group - A

1. Introduction to Bioprocess Engineering.
2. Bioreactors.
3. Isolation, preservation and maintenance of industrial microorganisms.
4. Kinetic of microbial growth and death.
5. Air and media sterilization.

Group - B

1. Media of industrial Fermentation.
2. Types of fermentation process: Analysis of batch, Fed-batch and continuous bioreactors, solid state fermentation, bio transformation.
3. Stability of microbial reactors, analysis of mixed microbial population, specialized bioreactors (pulsed, fluidized, photo-bioreactor etc.).
4. Measurement and control of bioprocess parameters.

Group - C

1. Downstream processing: Introduction, removal of microbial cells and solid matter, foam preparation, precipitation, filtration, centrifugation.
2. Cell disruption, liquid extraction, Chromatography, membrane process, drying and crystallization.
3. Enzyme and whole cell immobilization and their industrial applications.

Group - D

1. Industrial production of - alcohol (ethanol), acids (citric, acetic and gluconic), solvents (glycerol, acetone, butanol).
2. Industrial production of - Antibiotics (penicillin, streptomycin, tetracycline) amino acids (lysine, glutamic acid), single cell protein.
3. Introduction to food technology
 - Elementary idea of canning and packing.
 - Sterilization and pasteurization of food products.
 - Technology of typical food / food products (bread, cheese, idli)
 - Food preservation.



RKDF UNIVERSITY, RANCHI

M.Sc. (Bio-Technology)

Essential Readings

- Pappler, Microbial technology, Volume 1,2 7 3 Academicpress
- E.L. Mansi, Fermentation, Microbiology & Biotechnology, TaylorPub.
- Murray Moo & Young, Comprehensive Biotechnology, Vol-1 to4.
- Tripathi, Food Biotechnology, DominantPublication
- Mukhopadhyay. Process Biotechnology Fundamental. Vivabook
- Shuler and Kargi, Bioprocess engineering. Prentice-Hall.
- Schugertl. 1987. Bioreactionengineering.J/W.
- Stanbury and Whitaker. Principles of fermentationtechnology.
- Sikyta, Methods in Industrial microbiology. Ellis HardwoodLtd.
- T.K.Ghose. Bioprocess computation in biotechnology. Ellis HardwoodLtd.
- Murray Joh. 1997. Microorganisms andBiotechnology.
- Bioprocess Engineering Principles by Doran (D); Academic Press,1998
- Cooney, A.E. Humphrey, Comprehensive Biotechnology: The principles & Regulation of Biotechnology in Industry, Agriculture and Medicine, Vol.2, PergamonPress.
- Doran. Bioprocess Engineering Principles - Academic Press -2001



Branch	Subject Title	Subject Code
M.Sc. Biotechnology	Practical – I Lab	PBT155

1. Method of Protein Estimation
 - i) Estimation of Protein by Biuret methods
 - ii) Estimation of Protein by Folin Lowry methods
 - iii) Estimation of Protein by Bradford method
 - iv) Estimation of Protein by UV Absorption.
2. Method of Carbohydrate Estimation
 - i) Estimation of reducing sugar by DNSA method.
 - ii) Estimation of Carbohydrate by Nelson-Somogyi method
 - iii) Estimation of Carbohydrate by GOD/POD method.
 - iv) Estimation of Carbohydrate by Phenol Sulphuric acid Method.
3. Nucleic acid Estimation
 - i) Estimation of DNA by DP method
 - ii) Estimation of RNA by orcinol method / modified orcinol
 - iii) Estimation of total lipids in seeds
4. Analysis of oils, iodine numbers, saponification value, acid number
5. Enzyme assay, Enzyme Kinetics, specific activity, Determination of K_m & V_{max} , Optimum pH, Optimum Temperature of Amylase / Alkaline phosphatase / protease / cellulase
6. Studying comparative effect of Inhibitors on enzyme activity of Amylase/Alkaline phosphatase / protease / cellulose.

Alkaline Phosphatase i.e., a) Competitive Inhibition (NaH_2PO_4 , PNP)
 b) Uncompetitive Inhibition (L – Phenylalanine)
7. Separation of plant pigments by paper chromatography
8. Separation of Amino acids by thin layer chromatography
9. Cell motility and flagella staining
10. Isolation of chlorophyll and xanthophyll from spinach leaves

1. Isolation & maintenance of organism by plating, streaking & serial isolation methods slants & stab culture, storage of microorganism
2. Microscopic observation - Gram staining, Capsule & Spore Staining



RKDF UNIVERSITY, RANCHI

M.Sc. (Bio-Technology)

3. Growth curve –Diauxic
 4. Effect of Environmental Factors on Growth of Bacteria: Salt, Temp,pH.
 5. Viable count of bacteria from soil sample (Dilution PlatingMethod)
 6. Biochemical characterization of selectedMicrobes
 7. Isolation of bacteriophages from sewage sample
 8. Enrichment and Isolationof:
 - a) Halophiles b) Acidophiles c) Phenol Degraders
 - d) Nitrogen Fixers e) Antibiotic Producers f) Kojic Acid Producers
 9. AlcoholFermentation
 10. Comparative studies of ethanol production using different substrates
 11. Immobilization of WholeCells
 12. Effect of Antibiotics on various Gram Positive and Gram Negativebacteria
 13. Determination of Minimum Inhibitory Concentration (MIC) andMinimum
 14. Bactericidal Concentration (MBC) of various Antibiotics on differentOrganisms
 15. Biochemical tests for identification of Bacteria- Oxidase, Catalase, IMVIC test, TSI Testetc.
 16. Growth curve, measure of bacterial population by turbidometry and studying the effect of temp., pH Carbon andNitrogen
 17. Isolation of Rhizobium sp. From Root nodules oflegumes.
- III MolecularBiology
1. Isolation of Genomic DNA from bacterial cell / plantcell
 2. Isolation of RNA from Yeastcells
 3. Determination of T_m values ofDNA
 4. Isolation of Temperature sensitive conditionalMutant
 5. Isolation of auxotrophic mutant by 5 BrUmutagenesis
 6. BacterialConjunction
 7. Physical mapping with interrupted conjugation techniques (By Problem solvingapproach)
 8. BacterialTransformation
- IV MolecularTechniques
1. Isolation & Characterization of plasmidDNA
 2. Isolation of Lambda phageDNA
 3. Quantification of nucleicacid
 4. Cloning in Plasmid or Phagevectors
 5. SouthernBlotting
 6. Development of RFLP & RAPD Map
 7. Access of population diversity by 16S rRNAsequence.



RKDF UNIVERSITY, RANCHI

M.Sc. (Bio-Technology)

SCHEME SEMESTER – II

S.No.	Subject Code	Subject Name	Marks Distribution				
			Internal	External		Total	
				Max	Min	Max	Min
1.	PBT201	Genetic Engineering	30	70	21	100	35
2.	PBT202	Biochemistry & Biophysics	30	70	21	100	35
3.	PBT203	Immunology	30	70	21	100	35
4.	PBT204	Environmental Biotechnology	30	70	21	100	35
Total						400	
<i>Practical</i>							
S.No.	Code	Subject Name	Max		Min		
1.	PBT255	Practical - II Lab	100		50		



RKDF UNIVERSITY, RANCHI

M.Sc. (Bio-Technology)

Choice Based Credit System Semester - II

S.No.	Subject Code	Subject Name	L	T	P	Credit
1.	PBT201	Genetic Engineering	3	1	0	4
2.	PBT202	Biochemistry & Biophysics	3	1	0	4
3.	PBT203	Immunology	3	1	0	4
4.	PBT204	Environmental Biotechnology	3	1	0	4
5.	PBT255	Practical - II Lab	0	0	8	4
Total			12	4	8	20



RKDF UNIVERSITY, RANCHI

M.Sc. (Bio-Technology)

SEMESTER - II

Branch	Subject Title	Subject Code
M.Sc. Biotechnology	Genetic Engineering	PBT201

Group - A

1. Scope of genetic engineering.
2. Milestone in genetic engineering: DNA sequencing, synthesis and mutation, detection and separation, cloning, gene expression, cloning and patenting of life forms, genetic engineering guidelines.
3. Molecular tools and their applications: Restriction enzymes, modification enzymes, DNA and RNA markers. Molecular Markers- RFLP maps, RAPD, STS, microsatellites SCAR (sequence characterized Amplified Regions) SSCP (single stranded conformational polymorphism), AFLP, QTL, application of molecular marker.
4. Isolation of genomic and plasmid DNA and their purification.
5. PCR and Its Applications: Primer design; Fidelity of thermostable enzymes; DNA polymerases; Types of PCR – multiplex, nested, reverse transcriptase, real time PCR, touchdown PCR, hot start PCR, colony PCR, cloning of PCR products; T vectors; Proof reading enzymes; PCR in gene recombination; Deletion; addition; Overlap extension; and SOEing; Site specific mutagenesis; PCR in molecular diagnostics; Viral and bacterial detection; PCR based mutagenesis, Mutation detection: SSCP, DGGE, RFLP, Oligo Ligation Assay (OLA), MCC (Mismatch Chemical Cleavage, ASA (Allele-Specific Amplification), PTT (Protein Truncation Test)

Group - B

1. Gene cloning vectors: Plasmids, bacteriophages, phagemids, cosmids, Artificial chromosomes.
2. Restriction mapping of DNA fragment and map construction, nucleic acid sequencing.
3. cDNA synthesis and cloning ; reverse transcription, linkers, adaptor and their chemical synthesis, probes, library construction and screening.
4. Alternative strategies of gene cloning: cloning interacting genes - two-and-three hybrid systems, cloning differentially expressed genes, nucleic acid microarray.



RKDF UNIVERSITY, RANCHI

M.Sc. (Bio-Technology)

Group - C

1. Site-directed mutagenesis and protein engineering. Proteomics Protein analysis (includes measurement of concentration, amino-acid composition, N-terminal sequencing); 2-D electrophoresis of proteins; Microscale solution isoelectric focusing; Peptide fingerprinting; LC/MS-MS for identification of proteins and modified proteins; MALDI-TOF; SAGE and Differential display proteomics, Protein-protein interactions, Yeast two hybridsystem.
2. How to study gene regulation? DNA transaction, Northern blot, primer extension, SI mapping, RNase protection assay, Reporterassays.
3. Expression strategies for herterologous genes: expression in bacteria, expression in yeast, expression in insects and insect cells, expression in mammalian cells, expression inplants.

Group - D

1. T- DNA and transposon tagging: Role of gene tagging in gene analysis, T-DNA and transposon tagging, Identification and isolation of genes through T-DNA ortransposon.
2. Transgenic and gene knockout technologies: Targeted gene replacement chromosome engineering. Genetherapy.
3. Pharmacogenetics: High throughput screening in genome for drug discovery- identification of gene targets, Pharmacogenetics and drugdevelopment

Essential Readings

- Nicholl, An Introduction to Genetic Engg. - 2 ed,Cambridge
 - Primrose, Principles of Gene Manipulation - 6 ed,Blackwell
 - Winnacker, From Genes to Clones,Panima
 - Primrose, Principle of Gene Manipulation,Blackwell
 - Griffiths, Intro. to Genetic Analysis - 8 ed, FreemanPub.
 - Maxine singer-berg, Genes – Genomes, Uni. Sci.Book
 - T.A.Brown, Gene Clonning - DNA Analysis,Blackwell
 - John Witkowski, Recombinant DNA, Scientific American
 - Piramal, Molecular Biotechnology, DominantPub.
 - Maxine singer & Paul Berg, Exploring Genetic Mechanism, Uni.scienceBooks
 - Bruce K. Patterson, Techniques in Quantification and Localization of Gene Expression, BirkhausPub.
 - Reed ,Holmes,Jonathan Weyers, Practical Skills in BimolecularSciences
 - Anthony, Griffiths,William M, Modern Genetic Analysis :Integrating Genes & Genomes, W.H.Freemanand Company
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RKDF UNIVERSITY, RANCHI

M.Sc. (Bio-Technology)

SEMESTER - II

Branch	Subject Title	Subject Code
M.Sc. Biotechnology	Biochemistry & Biophysics	PBT202

Group - A

1. Polysaccharides- Types, structural features, methods for compositional, analysis.
2. Carbohydrates: Metabolism of carbohydrates, Glycolysis, Krebs cycle, hexose Monophosphate Pathway, Glycogenesis, Gluconeogenesis, Uronic Acid Pathway, Glyoxylate Cycle, Calvin Cycle, Hatch-Slack Pathway, Light reaction and formation of high energy bonds.
3. Sugars- Classification and reactions.
 - a. Lipids- Classification, structure and functions.
 - b. Lipid/ fat metabolism: Occurrence of various fatty acids in nature; biosynthesis of saturated and unsaturated fatty acids, sterols, phospholipids, triglycerides; biodegradation of saturated and unsaturated fatty acids.

Group - B

1. Amino acids and peptides- Classification, chemical reaction and physical properties.
2. Proteins- Classification and separation, purification and criteria of homogeneity, Ramachandran map; Primary and secondary structure, alpha -helix, beta -sheet structure etc. tertiary and quaternary structure, structural features of membrane proteins, secondary and tertiary structure prediction of protein conformation, sequencing of proteins.
3. Glyco- and lipoprotein- Structure and function.

Group - C

1. Enzymes: IUB classification, Steady state kinetics of enzyme catalyzed single substrate reaction, factors affecting enzymes activity, Introduction to allosteric enzymes. Enzymes in food processing, medicines and diagnostics; enzyme immobilization (method of immobilization industrial and therapeutically uses immobilized enzymes); role of enzyme in ELISA.
2. Bioenergetics-basic principles; Equilibria and concept of free energy; Coupled processes; Glycolytic pathway; Krebs's cycle; Oxidative phosphorylation; Photosynthesis; Elucidation of metabolic pathways; Logic and integration of central metabolism; entry/ exit of various biomolecules from central pathways; Principles of metabolic regulation; Regulatory steps; Signals and second messengers.



RKDF UNIVERSITY, RANCHI

M.Sc. (Bio-Technology)

Group - D

1. Bio-techniques Basic Techniques Buffers; Methods of cell disintegration; Enzyme assays and controls; Detergents and membrane proteins; Dialysis, Ultrafiltration and other membrane techniques
2. Spectroscopy Techniques: UV, Visible and Raman Spectroscopy; Theory and application of Circular Dichroism; Fluorescence; MS, NMR, PMR, ESR and Plasma Emission spectroscopy
3. Chromatography Techniques: TLC and Paper chromatography; Chromatographic methods for macromolecule separation - Gel permeation, Ion exchange, Hydrophobic, Reverse-phase and Affinity chromatography; HPLC and FPLC; Criteria of protein purity
4. Electrophoretic techniques: Theory and application of Polyacrylamide and Agarose gel electrophoresis; Capillary electrophoresis; 2D Electrophoresis; Disc gel electrophoresis; Gradient electrophoresis; Pulsed field gel electrophoresis.
5. Centrifugation: Basic principles; Mathematics & theory (RCF, Sedimentation coefficient etc); Types of centrifuge - Microcentrifuge, High speed & Ultracentrifuges; Preparative centrifugation; Differential & density gradient centrifugation; Applications (Isolation of cell components); Analytical centrifugation; Determination of molecular weight by sedimentation velocity & sedimentation equilibrium methods
6. Radioactivity Radioactive & stable isotopes; Pattern and rate of radioactive decay; Units of radioactivity; Measurement of radioactivity; Geiger-Muller counter; Solid & Liquid scintillation counters (Basic principle, instrumentation & technique); Brief idea of radiation dosimetry; Cerenkov radiation; Autoradiography; Measurement of stable isotopes; Falling drop method; Applications of isotopes in biochemistry; Radiotracer techniques; Distribution studies; Isotope dilution technique; Metabolic studies; Clinical application; Radioimmunoassay
7. Advanced Techniques: Protein crystallization; Theory and methods; API-electrospray and MALDI-TOF; Mass spectrometry; Enzyme and cell immobilization techniques; DNA & Peptide Synthesis.

Essential Readings

- Lenhinger. Principles of Biochemistry, Nelson & Cox, 4th Edition.
- Stryer – Biochemistry. W.H. Freeman & Co.
- Plumner. An introduction to practical Biochemistry, 3rd Edition
- J. Jayaraman. Lab Manual in Biochemistry.
- Cohn and Stumph. Outline of Biochemistry. Wiley Eastern.
- Zube's Biochemistry. 4th Edition Macmillan.
- Switzer and Garritty. Experimental Biochemistry WH Freeman. 2nd Edition



RKDF UNIVERSITY, RANCHI

M.Sc. (Bio-Technology)

- Voet&Voet Donald. 3rd Edition. Fundamentals of Biochemistry,J/W.
 - Hames and Hooper. 2000. Instant notes in Biochemistry. BIOS Sci.Publ.
 - Smith G. 1996. Biotechnology. Cambridge Univ.Press.
 - Geoffrey Cooper. 2000. The cell with CD- Rom. Sinauer Asso.Incorp.
 - Elliott & Elliot.3rd Edition Biochemistry and molecularbiology.
 - Seidman and Moore. 2000. Basic laboratory methods for biotechnology.Longman
 - Boyer. 1999. Concepts in biochemistry.Thomson
 - Das and Mookerjee. Outline ofbiology.
 - Biotechnology, Demystifying the concepts. By David Bourgaize. Alp2000
 - Wilson, & Walker. 1995. Principles and techniques of practicalBiochemistry.
 - Boyer. 2001. Concepts in Biochemistry. 2ndEdition
 - Hames, Instant Notes in biochemistry, 2ndEdition.
 - Garrett, Biochemistry,2ndEdition.
 - Price & Steven, Fundamentals of Enzymology,3rdEdition
 - Creighton, proteins: Structure & Molecular Properties, FreemanPub.+
 - Stephen Neidle, Nucleic acid Structure and Recognition, Oxford UniversityPress
 - Rob Reed , David Holmes, Practical Skills in Bimolecular Sciences,
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RKDF UNIVERSITY, RANCHI

M.Sc. (Bio-Technology)

SEMESTER - II

Branch	Subject Title	Subject Code
M.Sc. Biotechnology	Immunology	PBT203

Group - A

- (a) Introduction
 1. Phylogeny of immune system
 2. Innate and acquired immunity
 3. Clonal nature and immune response
- (b) Organization and structure of lymphoid organs.
- (c) Nature and biology of antigens and super-antigens.
- (d) Antibody structure and function.
- (e) Antigen-antibody interactions.
- (f) BCR & TCR generation of diversity.
- (g) Cells of the Immune System: Haematopoiesis and differentiation, Lymphocytes trafficking, B-lymphocytes, T-lymphocytes, Macrophages, Dendritic cells, natural killer and Lymphokine activated killer cells, Eosinophil, Neutrophils and Mast cells.
- (h) Regulation of immune response: Antigen processing and presentation, generation of humoral and cell mediated immune responses; Activated B and T-lymphocytes; cytokine and their role in immune regulation; T-cell regulation, MHC-restriction.
- (i) Cell-mediated cytotoxicity: Mechanism of T-cell and NK cell mediated lysis
Antibody dependent cell mediated cytotoxicity, macrophage mediated cytotoxicity.

Group - B

1. Major histocompatibility complex.
2. Complement system.
3. Immunological tolerance.
4. Hypersensitivity.
5. Autoimmunity.
6. Transplantation.



RKDF UNIVERSITY, RANCHI

M.Sc. (Bio-Technology)

Essential Readings

- Janis Kuby, Immunology, 5th Edition
 - Ivan Roitt, Essential Immunology, 9th Edn.
 - Ananthnarayan, Medical microbiology,
 - Mary S. Leffell, & Noel R. Rose, Handbook of Human Immunology, CRCpress
 - Tizzard, Immunology
 - Elger, Immunology
 - Lidyard, Instant notes in Immunology, 2nd Edition.
 - Darla J wise, Immunology-A comprehensive review : A Blackwell science Pub.
 - Todd & Spickett, Immunology
 - Delves & Roitt Encyclopedia of Immunology- Vol-1 to Vol.-4, 2nd Edition
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RKDF UNIVERSITY, RANCHI

M.Sc. (Bio-Technology)

SEMESTER - II

Branch	Subject Title	Subject Code
M.Sc. Biotechnology	Environmental Biotechnology	PBT204

Group - A

1. Environment: basic concepts and issues.
2. Environmental pollution
3. Global environmental problems; ozone depletion, UV-B, green house effect and acid rain their impact and biotechnological approaches for management.
4. Air pollution and its control through biotechnology.

Group - B

1. Water pollution and its control: water as a scarce natural resource, need for water management, measurement of water pollution, sources of water pollution, waste water collection, waste water treatment- physical, chemical and biological treatment processes.
2. Microbiology of water treatment, aerobic process: activated sludge, oxidation ditches, trickling filter, towers, rotating disc, rotating drums and oxidation ponds.
3. Anaerobic process: anaerobic digestion, anaerobic filters, upflow anaerobic sludge.
4. Treatment schemes for waste water of dairy, distillery, tannery, sugar and antibiotic industries.

Group - C

1. Microbiology of degradation of xenobiotics in environment- ecological consideration, decay behaviour and degradative plasmids, hydrocarbons, substituted hydrocarbons, hydrocarbon transformation, oil pollution, surfactants, pesticides.
2. Bioremediation of contaminated soil and waste land.
3. Biopesticides in integrated pest management.
4. Solid wastes sources and management (Composting, vermin culture and methane production)

Group - D

1. Types of IP: Patents, Trademarks, Copyright & Related Rights, Industrial Design, Traditional Knowledge, Geographical Indications, Protection of New GMOs; International framework for the protection of IP
2. IPR Patent application-; Types of patent applications: provisional and complete specifications; PCT and convention patent applications; International patenting- requirement, procedures and costs Patent infringement- meaning, scope, litigation,



RKDF UNIVERSITY, RANCHI

M.Sc. (Bio-Technology)

case studies and examples

3. Bio-safety: Introduction; Historical Background; Introduction to Biological Safety Cabinets; Primary Containment for Biohazards; Biosafety Levels Biosafety guidelines - Government of India;
4. Definition of GMOs & LMOs Environmental release of GMOs; Risk Analysis; Risk Assessment; Risk management and communication; Cartagena Protocol., Bio-piracy.

Essential Readings

- Rajvaidhya, Environmental Biochemistry, APHPub
- Ahmed, Industrial & Envi. Biotech, Horizon
- Bitton, Wastewater Microbiology - 2 ed, Wiley
- D.P. Singh, Environmental Microbiotech, NewAge
- Pratham Vashisth, Environmental Biotechnology, DominantPub.
- Arihat Parulkar, Environmental Biotechnology, DominantPub
- Purohit Shammi, Environmental Sciences, StudentEdi
- Eugene Odum, Ecology, Oxford
- B.K. Singh, Biodiversity, MangalDeep
- 10. Vashishtha, Environmental Biotechnology, DominantPub.
- 11. Gerba & Pepler, Environment Microbiology
- H. D. Kumar, Textbook of Microbiology

Important Links

- <http://www.w3.org/IPR/>
- <http://www.wipo.int/portal/index.html.en>
- http://www.ipr.co.uk/IP_conventions/patent_cooperation_treaty.html www.patentoffice.nic.in
- www.iprlawindia.org/ - 31k - Cached - Similarpage
- <http://www.cbd.int/biosafety/background.shtml>
- <http://www.cdc.gov/OD/ohs/symp5/jyrtext.htm>
- <http://web.princeton.edu/sites/ehs/biosafety/biosafetypage/section3.html>



RKDF UNIVERSITY, RANCHI

M.Sc. (Bio-Technology)

SEMESTER - II

Branch	Subject Title	Subject Code
M.Sc. Biotechnology	Practical – II Lab	PBT255

I. Biochemistry and Biophysics

1. Protein Purification Studies of different proteins/enzymes.
2. SDS PAGE and Native Gel
 - a. CBB – R250 staining technique
 - b. Silver staining technique
2. Gel Filtration Chromatography
3. Ion Exchange Chromatography: Purification of proteins/enzymes using CM
 - a. Cellulose / DEAE Cellulose.
 - b. Induction of Protein synthesis in E.coli cells.
 - c. Determination of T_m value of nucleic acid
4. Determination of % G+C content
5. The ultraviolet absorption of proteins and amino acids
6. To determine an unknown protein concentration by plotting a standard graph of BSA using UV-Vis- spectrophotometer and validating the Beer-Lambert's law
7. Protein Folding Studies.

II. Genetic Engineering

1. Plasmid Curing by Acridine Orange
2. Restriction Digestion of λ DNA using three Restriction Endonuclease enzymes:
 - a) EcoR V b) Hind III c) BamH I
3. Replica plating techniques
4. Agarose gel electrophoresis and restriction mapping of DNA
5. Demonstration of techniques of PCR

III. Immunology

1. ELISA
 2. Generation of primary antibody by using mice as model organism.
 3. Separation of serum and plasma from whole blood.
 4. Single Radial Immunology
 5. Ouchterlony Double Diffusion.
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RKDF UNIVERSITY, RANCHI

M.Sc. (Bio-Technology)

SCHEME SEMESTER – III

S.No.	Subject Code	Subject Name	Marks Distribution				
			Internal	External		Total	
				Max	Min	Max	Min
1.	PBT301	Biostatistics, Computer Applications and Bioinformatics	30	70	21	100	35
2.	PBT302	Plant Biotechnology	30	70	21	100	35
3.	PBT303	Animal Biotechnology	30	70	21	100	35
4.	PBT304	Agricultural Microbiology	30	70	21	100	35
Total						400	
<i>Practical</i>							
S.No.	Code	Subject Name	Max		Min		
1.	PBT355	Practical - III Lab	100		50		



RKDF UNIVERSITY, RANCHI

M.Sc. (Bio-Technology)

Choice Based Credit System Semester - III

S.No.	Subject Code	Subject Name	L	T	P	Credit
1.	PBT301	Biostatistics, Computer Applications and Bioinformatics	3	1	0	4
2.	PBT302	Plant Biotechnology	3	1	0	4
3.	PBT303	Animal Biotechnology	3	1	0	4
4.	PBT304	Agricultural Microbiology	3	1	0	4
5.	PBT355	Practical - III Lab	0	0	8	4
Total			12	4	8	20



RKDF UNIVERSITY, RANCHI

M.Sc. (Bio-Technology)

SEMESTER - III

Branch	Subject Title	Subject Code
M.Sc. Biotechnology	Biostatistics, Computer Applications and Bioinformatics	PBT301

Group - A

1. Brief description and tabulation of data and its graphical representation.
2. Measure of central tendency and dispersion: Mean median, mode, range, standard deviation, and variance. Idea of two types of errors and level of significance, tests of significance (F & t test); chi - square tests.
3. Simple linear regression and correlation.

Group - B

1. Introduction of digital computers: Organization; low-level and high-level languages, binary number system.
2. Flow charts and programming techniques.

Group - C

1. Introduction to data structures and database concepts, introduction to Internet and its applications.
2. Introduction to MS-Office software, covering Word Processing, Spreadsheets and Presentation software, Introduction to Hardware graphics/ Coreldraw.

Group - D

1. Computer- Oriented Statistical Techniques: Frequency table of single discrete variable, Bubble sort, Computation of mean variance and standard deviation: t-test, correlation coefficient.
2. Introduction to Bioinformatics: Definition and aims; fundamental of data base searching (BLAST and FASTA); computational gene finding-multiple alignment and sequence search; molecular evolution and phylogenetic



RKDF UNIVERSITY, RANCHI

M.Sc. (Bio-Technology)

Essential Reading

- Gary B. Fogel & David Corne, Evolutionary Computation in Bioinformatics, Morgan Kaufmann Publishers,
- Christoph W. Sensen, Essentials Of Genomics and Bioinformatics, Wiley-VCH.
- David Bowtell and Joseph Sambrook, DNA Microarrays, CSHL Press
- Murray R. Selwyn, Principles of Experimental Design for the Life Sciences, CRC Press
- Warren J. Ewens, Gregory R. Grant, Statistical Methods in Bioinformatics : An Introduction, Springer.
- Hugh G. Griffin, Annette M. Griffin, PCR Technology: current innovation.
- Ben Hui Liu, Statistical Genomics : Linkage, mapping and QTL Analysis, CRC Press
- Westhead, Instant Notes on Bioinformatics, 1st Edn.
- Baxevanis & Francis, Bioinformatics, 2nd Edn. Wiley.
- T.K. Attwood, An Introduction to Bioinformatics, Pearson Pub.
- Higgs & Attwood, Bioinformatics & Molecular Evolution
- David W. Mount, Bioinformatics : Sequence and Genome Analysis, Cold Spring Harbor Laboratory Press



RKDF UNIVERSITY, RANCHI

M.Sc. (Bio-Technology)

SEMESTER - III

Branch	Subject Title	Subject Code
M.Sc. Biotechnology	Plant Biotechnology	PBT302

Group - A

1. History of plant tissue culture and its present status.
2. Introduction to Cell and tissue culture technique to produce novel plant and hybrids.
3. Tissue culture media (composition and preparation)
4. Phytohormones a) Chemical nature, biosynthesis, physiological roles and mode of action of Auxins, Gibberellins and Cytokinnins b) Chemical nature and physiological roles of Morphactin, Absciscic acid and Ethylene.

Group – B

1. Initiation and maintenance of callus and suspension culture; single cell clones.
2. Organogenesis, somatic embryogenesis; transfer and establishment of whole plants in soil.
3. Shoot- tip culture: Rapid clonal propagation and production of virus -free plants.
4. In vitro pollination, Embryo culture, embryo rescue and synthetic seeds.

Group - C

1. Anther, Pollen and Ovary culture for production of haploid plants and homozygous lines; endosperm culture.
2. Protoplast isolation, culture and fusion: selection of hybrid cells and regeneration of hybrid plants; symmetric and asymmetric hybrids, cybrids.
3. Nuclear cytology of cultured plant cells and somaclonal variation.
4. Cryopreservation, slow growth and DNA banking for germplasm conservation.

Group - D

1. Plant transformation technology; basis of tumor formation, mechanism of DNA transfer, role of virulence genes, use of Ti and Ri plasmids as vectors, co-integrative and binary vectors, use of reporter genes, particle bombardment, electroporation, microinjection transformation of monocots.
2. Transgenic plants: insect resistance, virus resistance, resistance to fungal and bacterial diseases, longer shelf life, male sterility.
3. Metabolic engineering and industrial products: Important plant secondary metabolites, control mechanisms and manipulation.
4. Concept of Phyto immunity.



RKDF UNIVERSITY, RANCHI

M.Sc. (Bio-Technology)

SEMESTER - III

Branch	Subject Title	Subject Code
M.Sc. Biotechnology	Animal Biotechnology	PBT303

Group - A

1. Structure and organization of animal cell
2. Equipment and materials for animal cell culture technology
3. Primary and established cell line culture
4. Introduction to the balanced salt solutions and simple growth medium, Brief discussion on the chemical, physical and metabolic functions of different constituents of culture medium, role of carbon dioxide, role of serum and supplements.
5. Serum and protein free defined media and their application.
6. Biology and characterization of the cultured cell, measuring parameters of growth.
7. Basic techniques of mammalian cell culture in vitro; disaggregation of tissue and primary culture; maintenance of cell culture; cell separation.
8. Scaling - up of Animal cell culture.
9. Cell synchronization.

Group - B

1. Cell cloning and micromanipulation.
2. Cell transformation.
3. Application of animal cell culture - special secondary metabolites/products (insulin, Human Growth Hormones, Interferons, t-plasminogen activator, Factor VIII etc.), Hybridoma technology.
4. Stem cell cultures, embryonic stem cells and their applications
5. Cell culture based vaccines
6. Transgenic animal

Essential Readings

- Freshney, Culture of animal Cells, Sixth Edition



RKDF UNIVERSITY, RANCHI

M.Sc. (Bio-Technology)

SEMESTER - III

Branch	Subject Title	Subject Code
M.Sc. Biotechnology	Agricultural Microbiology	PBT304

UNIT – I

History, scope and development of agricultural microbiology, rhizosphere and phyllosphere: concept, importance, factors affecting microbial diversity.

UNIT – II

Soil health: crop residues, humus, mineralization, immobilization, soil-sickness, composting, vermicomposting, green manure. Effect of crop residues on plant growth; biodegradation of pesticides and pollutants; biodegradation fate, bioavailability, acceleration, bioremediation. Biofertilizers: types, production, formulation and constraints.

UNIT – III

General idea about major agricultural pests: Plant diseases- late blight potato, downy mildew of pea, stem gall of coriander, powdery mildew / rust / smut, rust of linseed, Ergot of bajara, Anthracnose of soybean, Tikka disease of groundnut, wilt of arhar, bacterial blight of paddy, citrus canker, leaf curl of papaya, little leaf of brinjal. Insects: gram, soybean. Weeds: parthenium, xanthium, waterhyacinth, cyperus, phalaris

UNIT – IV

Post harvest losses of agricultural products: causes, problems and management recent trends in pest management: strategies, mass production, formulation and application technology, achievements, constraints

UNIT – V

Biotechnology in agriculture: the new green revolution, transgenic crops, gene protection technology, frost control technology, resistant varieties. Bioconversion futurology: exploitation of agricultural wastes for food / feed and fuel.

List of Recommended Books

- Soil microbiology by Subba Rao
- Soil and microbes by Waksman and Starkey.
- Plant pathology by Mehrotra.
- Alexander, M. Introduction to Soil Microbiology, 3rd Edition. Wiley Eastern Ltd., New Delhi
- Microbiology by S.S. Purohit.



RKDF UNIVERSITY, RANCHI

M.Sc. (Bio-Technology)

SEMESTER - III

Branch	Subject Title	Subject Code
M.Sc. Biotechnology	Practical - III Lab	PBT355

1. Use of NCBI Bioinformatics Tools:
a) Pubmed b) OMIM c) Taxonomy
d) Protein analysis e) Genes and Disease
2. Use of Expasy Tools
3. FASTA and sequence formats
4. BLAST
5. Alignments – Pair wise & global
6. Construction of Dendrogram
7. Prediction of ORF/ gene Prediction
8. Protein Visualization (RASMOL, SPDB VIEWER, PROTEINEXPLORER)
9. Protein Modeling-Homology modeling & Active site Prediction
10. Primer Designing
11. Biostatistics: Mean, Median, Mode, Standard Deviation, Chi square test, Student's t-test

II. Plant Biotechnology

1. Isolation of chloroplast & estimation of chlorophyll
2. Preparation of media & Surface sterilization of Explant
3. Selection, preparation and inoculation of explant for callusing
4. Study of callus characteristics
5. Sub-culturing of callus in differentiation media
6. Sub-culturing callus for Suspension culture
7. Extraction of secondary metabolites from callus culture
8. Protoplast Isolation & Culture
9. Production of Haploids by anther culture
10. Agro bacterium mediated gene transfer and reporter gene assay
11. In vitro evaluation of Medicinal plants against pathogenic microbes
12. Role of Microorganisms in elevation of heavy metal induced stress in plants



RKDF UNIVERSITY, RANCHI

M.Sc. (Bio-Technology)

III. Animal Biotechnology

1. Total count of RBC & WBC differential count & Blood grouping
2. Western Blotting
3. Isolation & staining of Mitochondria
4. Blood film preparation and identification of cells.
5. Demonstration of Immunological reaction (WIDAL, VDRL Pregnancy, Hepatitis)

IV. Agricultural Microbiology

1. Isolation and Enumeration of the microorganism from soil by serial dilution agar plate method.
2. Isolation of fungi from soil by warcup's method.
3. Isolation of azotobacter species from soil.
4. Isolation of microorganism from rhizosphere.
5. Isolation of microorganism from phyllosphere (phyloplane) by serial dilution, agar plate method or leaf impression method.



RKDF UNIVERSITY, RANCHI

M.Sc. (Bio-Technology)

SEMESTER - IV

Branch	Subject Title	Subject Code	Marks Distribution		Credit
M.Sc. Biotechnology	Dissertation	PBT466	Max 500	Min 250	20

Overall project dissertation may be evaluated under the following heads:

- Motivation for the choice of topic
- Project dissertation design
- Methodology and Content depth
- Results and Discussion
- Future Scope & References
- Presentation style
- Viva-voce

PROJECT WORK/ DISSERTATION/ PAPER PRESENTATION

Each student has to submit two copies of the dissertation work duly forwarded by the HOD of Department concerned. The forwarded copies will be submitted in the Department of Biotechnology, RKDF University, Ranchi for evaluation (Seven days before the seminar).

The paper will consist of

(a) Field work/Lab work related to the project.

(b) Preparation of dissertation based on the work undertaken.

(c) Presentation of project work in the seminar on the assigned topic in the P.G.

Department of Biotechnology, RKDF University, Ranchi & open viva thereon.

- Student alone or in a group of not more than five, shall undertake one Project approved by

the Subject Teacher/H.O.D. of the Department/College concerned. The progress of the Project shall be monitored by the faculty members at regular intervals.

OR

- Paper presentation on 'Topic Provided' and group discussion.